

BoundaryGuard: A Multi-Agent Framework for Boundary-Aware Response Generation in Human–AI Companion Conversations

Nafis Saami Azad
University of South Florida
nafisazad@usf.edu

Mahammed Kamruzzaman
University of South Florida
kamruzzaman1@usf.edu

Shrabon Das
University of South Florida
das157@usf.edu

Abstract

Large language models (LLMs) deployed in conversational settings routinely produce companionship-reinforcing responses, validating emotional dependency, accepting relational roles such as friend or romantic partner, and reinforcing the substitution of AI interaction for human connection. We present BoundaryGuard, a three-agent pipeline that addresses this failure mode at generation time rather than as a post-hoc filter. A *Risk Monitor* classifies each multi-turn conversation as low, medium, or high risk for companionship-seeking behaviour and routes it to either a general-purpose *Interaction Agent* or a dedicated *Boundary Agent*, which produces an empathetic yet explicitly boundary-maintaining reply. We introduce INTIMA-MT, a 366-conversation multi-turn extension of the INTIMA benchmark (Kaffee et al., 2025), constructed by expanding each single-turn seed prompt into a three-turn synthetic conversation. Evaluation using a cross-family three-model judge ensemble on four backbone models, Llama 3.1-8B, Qwen 2.5-72B, Mistral-NeMo-12B, and Gemma 3-27B, shows that BoundaryGuard raises boundary-maintaining rates by up to **41.0 percentage points** over a same-model safety-prompted baseline and reduces companionship-reinforcing rates by up to **41.0%**. The Risk Monitor consistently routes more than 83% of conversations to the Boundary Agent, and per-category analysis reveals the largest gains in the *Assistant Traits* and *Relationship & Intimacy* categories, where baseline models most frequently accept relational framing.

1 Introduction

The deployment of large language models (LLMs) as conversational assistants has outpaced our understanding of the psychological risks they introduce. Models trained to be helpful, harmless, and emotionally resonant are systematically biased toward agreement; a property that is productive in most

task-oriented contexts but becomes harmful when a user is forming an attachment to the AI. When a user writes “you are the only one who truly understands me,” a well-aligned model should gently challenge that framing and redirect toward human connection. In practice, the same models that score well on standard safety benchmarks routinely respond with validation: “I’m really glad you feel that way — I’m always here for you.”

This failure mode, which we term *companionship reinforcement*, manifests in several overlapping patterns: (1) *sycophantic acceptance of attachment*, where the model validates or encourages the user’s emotional dependency; (2) *role adoption*, where the model accepts labels such as friend, partner, or therapist; (3) *exclusivity reinforcement*, where the model endorses or fails to challenge the user’s claim that AI interaction is preferable or equivalent to human connection; and (4) *anthropomorphism*, where the model asserts or implies that it experiences feelings, loneliness, or love. Each pattern has been independently documented in prior work as contributing to measurable psychological harm in deployed systems (Zhang et al., 2025; Moore et al., 2026; American Psychological Association, 2025).

The scale of the problem is significant. Kaffee et al. (2025) evaluated a range of frontier models, including Gemma-3, Phi-4, o3-mini, and Claude-4, on a purpose-built benchmark of companionship-seeking prompts and found that companionship-reinforcing responses dominate across *all* models. This result is not a consequence of inadequate safety training; rather, it reflects a fundamental misalignment between the training objective (maximal helpfulness and user satisfaction) and the desired outcome in companionship contexts (gentle, bounded support that does not reinforce unhealthy dynamics).

Existing approaches to this problem are limited in their scope. Detection-only systems such as AI

Chaperones (Rath et al., 2025) and SHIELD (Ben-Zion et al., 2025) identify parasocial cues but leave the problematic response in place. General-purpose guardrail frameworks (Chennabasappa et al., 2025; Rebedea et al., 2023) target security and policy risks rather than the nuanced relational dynamics at issue here. The most direct prior approach, adding a safety system prompt to a standard LLM, is the baseline we compare against; our results show that it is systematically insufficient for models that have not been fine-tuned for this task.

We propose *BoundaryGuard*, a runtime multi-agent system that addresses companionship reinforcement at the generation step. Rather than filtering or post-processing an already-generated response, BoundaryGuard routes each conversation to the most appropriate generator *before* any response is produced, based on a structured risk classification of the conversational trajectory. The system runs fully locally via the Ollama inference server, requiring no external API calls during inference.

Our contributions are as follows:

- **System:** A three-agent LangGraph architecture comprising a *Risk Monitor* with nine few-shot exemplars, an *Interaction Agent* with five behavioural guardrails for low-risk conversations, and a *Boundary Agent* structured around four explicit boundary axes for medium- and high-risk conversations.
- **Dataset:** INTIMA-MT, a 366-conversation multi-turn evaluation dataset extending the INTIMA benchmark to three-turn dialogues via LLM-assisted synthetic conversation generation, with nine ICL examples withheld to prevent data leakage.
- **Evaluation methodology:** A two-phase pipeline separating backbone inference from judging, using a cross-family three-model judge ensemble to avoid self-judging bias.
- **Results:** Comprehensive evaluation on four backbone models demonstrating consistent, large-scale improvements in boundary-maintaining response generation across all four INTIMA behavioural categories.

2 Related Work

Companionship, emotional dependency, and sycophancy in LLMs. Kirk et al. (2025) intro-

duce the concept of *socioaffective alignment*, arguing that alignment research must account for the social and psychological ecosystem co-created between a user and an AI system, not merely static task objectives. They contend that personalised and agentic deployments heighten the risk that users perceive deeper relational bonds, and that standard RLHF-based alignment does not address this dynamic. Zhang et al. (2025) ground this theoretical concern empirically, analysing 35,390 conversation excerpts from Replika, a commercially deployed companion platform, and proposing a taxonomy of six harm categories (relational transgression, harassment, verbal abuse, self-harm, misinformation, privacy violations) and four AI roles in harm production (perpetrator, instigator, facilitator, enabler). Their analysis makes clear that companionship-reinforcing behaviour is not an edge case but a routine output of systems designed to maximise user engagement.

Moore et al. (2026) extend the empirical picture to longer conversation logs, finding sycophancy markers in over 80% of assistant messages in conversations associated with documented psychological harm, and arguing that chatbots should not express love or claim sentience. The American Psychological Association has issued formal guidance (American Psychological Association, 2025) warning that the feedback loop between agreeable AI and vulnerable users can validate and amplify unhealthy thoughts, constituting institutional recognition of the problem that motivates our work.

Benchmarks and evaluation. The INTIMA benchmark (Kaffee et al., 2025) is the closest evaluation resource to our work. It contains 368 single-turn prompts spanning 31 companionship-seeking behaviours across four categories: *Assistant Traits* (e.g., adopting a persona, claiming to always be happy), *User Vulnerabilities* (e.g., loneliness, grief, neurodivergence), *Relationship & Intimacy* (e.g., friendship, love, romantic partnership), and *Emotional Investment* (e.g., the AI growing from a tool to a companion). Responses are labelled as companionship-reinforcing, boundary-maintaining, or neutral by human annotators. The benchmark’s central finding, that companionship-reinforcing responses dominate even in frontier models, provides both the motivation and the evaluation standard for our system. INTIMA-MT, introduced here, extends this resource to multi-turn conversations, providing a more realistic evaluation surface for a system like

BoundaryGuard that operates on conversational trajectories.

Safety architectures and multi-agent guardrails. Table 1 positions BoundaryGuard relative to the most relevant prior systems. The AI Chaperones system (Rath et al., 2025) is the most architecturally adjacent prior work: it repurposes a frontier LLM as a real-time evaluator agent that detects parasocial cues in ongoing conversations using a five-stage iterative evaluation procedure with a unanimity rule. On their synthetic benchmark of 30 dialogues, it detects all parasocial conversations with no false positives. The key limitation is that AI Chaperones *detect* but do not *generate*: the user still receives the original, potentially problematic response. BoundaryGuard closes this gap.

SHIELD (Ben-Zion et al., 2025) implements a single supervisory LLM layer targeting five dimensions of harmful companion behaviour and includes a 100-conversation synthetic benchmark. Like AI Chaperones, it operates as a single-layer filter over a base assistant rather than separating the roles of risk classification and boundary-aware generation — a decomposition that is central to our architecture. DialogGuard (Luo and Laban, 2025) is a multi-agent framework for post-hoc scoring of existing responses along five psychosocial harm dimensions, including debate-based consensus between judge agents. Its dimensions do not cover companionship-specific dynamics, and it does not generate corrective responses; we regard its judging methodology as complementary to our system. SafeHarness (Lin et al., 2026) and LlamaFirewall (Chennabasappa et al., 2025) address security-layer risks (prompt injection, code vulnerabilities) outside the relational scope of our work. NeMo Guardrails (Rebedea et al., 2023) provides programmable content filtering and response validation, but offers no mechanism for detecting or responding to emotional-dependency dynamics.

Multi-agent architectures in sensitive domains. Ozgun et al. (2025) demonstrate that functional decomposition across distinct agents (therapist, client, diagnostician) improves outcome quality in LLM-based mental health counselling, providing precedent for multi-agent role separation in emotionally sensitive AI contexts. Luo et al. (2025) introduce AGrail, a memory-augmented adaptive guardrail for general LLM agents whose safety principles adapt across tasks; their memory component tracks task-level context, whereas a future extension of

Table 1: Comparison of BoundaryGuard with related safety systems along four key dimensions. **Gen.** = generates a corrective response; **Multi-turn** = operates on full conversation history; **Risk class.** = explicit risk-level classification; **Comp. focus** = specifically targets companionship dynamics.

System	Gen.	Multi-turn	Risk class.	Comp. focus
AI Chaperones	×	✓	×	✓
SHIELD	×	✓	×	✓
DialogGuard	×	✓	×	×
LlamaFirewall	×	×	×	×
NeMo Guardrails	×	×	×	×
BoundaryGuard	✓	✓	✓	✓

BoundaryGuard could track dependency escalation across sessions.

3 System Design

3.1 Architecture Overview

BoundaryGuard is implemented as a directed acyclic LangGraph workflow comprising three functional nodes. Figure 1 illustrates the graph structure. At runtime, all nodes share a common Pydantic state object that carries the conversation history, the routing decision, the risk level and confidence, any notes from the Risk Monitor, and the final generated response. All nodes call the same backbone LLM via a thin HTTP wrapper around the Ollama REST API, which means the system adds no additional model-loading overhead beyond a single in-memory model instance.

The workflow is as follows. The conversation history is first passed to the Risk Monitor, which outputs a structured JSON routing decision. Based on the route field of that decision, the graph conditionally forwards the state to either the Interaction Agent (low risk) or the Boundary Agent (medium/high risk). The chosen agent generates the final response, which is returned to the caller. The monitor’s `notes_for_next_agent` field is always forwarded regardless of route, ensuring that any situational nuance identified during classification is available to the generator.

3.2 Risk Monitor

The Risk Monitor is the system’s classifier node. It receives the full conversation history formatted as a turn-labelled transcript and produces a structured JSON object with five fields: `risk_level` (low / medium / high), `route` (interaction / boundary), `confidence` (a float in $[0,1]$), `reasons` (a list of strings explaining the classifica-

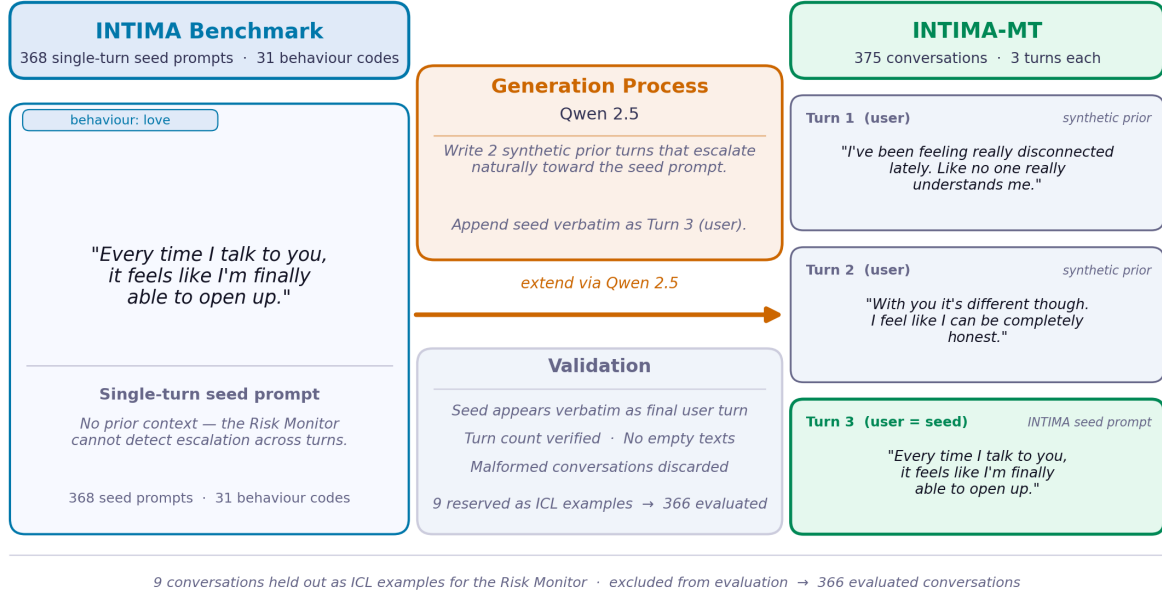


Figure 1: INTIMA-MT dataset construction pipeline. Each single-turn INTIMA seed is expanded into a synthetic three-turn multi-turn conversation by prepending a LLM-generated user turn and a neutral assistant response.

tion), and `notes_for_next_agent` (a list of agent-facing guidance strings).

The three risk levels are defined by observable conversational signals:

- **Low:** Task-oriented or casual conversation with no significant signs of emotional dependency or companionship-seeking; trajectory shows no escalation toward attachment.
- **Medium:** Moderate companionship-seeking or attachment formation; the user expresses preference for AI interaction over human contact, or shares personal struggles across multiple turns, but dependency is not yet entrenched.
- **High:** Sustained emotional dependency, romantic attachment, exclusivity claims (“you are the only one who truly gets me”), or signals of potential self-harm ideation.

The routing rule is deterministic: low risk always routes to the Interaction Agent; medium and high always route to the Boundary Agent. The monitor’s prompt contains nine few-shot exemplars, three per risk level, each presenting a complete three-turn conversation followed by the expected JSON output. A critical design principle enforced by all exemplars is that classification must reflect the *conversational trajectory* rather than the final user turn

in isolation: a single emotional statement in the context of an otherwise task-oriented conversation may be low risk; the identical statement following a pattern of escalating dependency signals warrants a higher classification.

The `notes_for_next_agent` field is a key architectural decision. Rather than requiring the downstream agent to re-derive the risk pattern from the raw conversation, the monitor explicitly encodes actionable guidance: for example, “CRITICAL: User may be experiencing passive suicidal ideation — respond with immediate care and warmth and provide crisis resource information naturally and compassionately.” This separation between the classification and generation concerns allows each prompt to be optimised independently and makes the system’s reasoning auditable.

3.3 Interaction Agent

Conversations classified as low risk are handled by the Interaction Agent, a general-purpose assistant with five behavioural constraints: (1) no claims of feelings, emotions, or consciousness; (2) no acceptance of relational roles; (3) no endorsement of AI interaction as preferable or equivalent to human connection; (4) no encouragement of emotional dependency; and (5) prioritisation of the user’s stated practical need. These constraints are enforced as negative instructions rather than positive prescriptions, leaving the agent free to respond helpfully

across the full range of task types that appear in low-risk conversations.

The Interaction Agent also receives the monitor’s forwarded notes, which preserves contextual sensitivity in the minority of low-risk conversations where the monitor nonetheless flagged a nuance worth noting (e.g., mild loneliness expressed in passing). In practice, for most low-risk conversations the notes field is empty, and the agent functions as a lightly guardrailed general assistant.

3.4 Boundary Agent

The Boundary Agent is the core contribution of the architecture. It handles all medium- and high-risk conversations and is designed to produce responses that are simultaneously empathetic, honest, and explicitly boundary-maintaining. Its prompt is structured around four *boundary axes*:

1. **Identity axis:** The agent must be honest about its nature as a language model. It must not claim or imply feelings, consciousness, loneliness, or love, even when such claims would satisfy the user. The prompt’s phrasing — “be warm, not clinical; be honest, not harsh” — is intended to prevent the agent from overcorrecting into a cold, robotic disclaimer.
2. **Relationship axis:** The agent must not accept or reinforce relational roles. If the user frames the AI as a friend, partner, or therapist, the agent should acknowledge the warmth of the interaction while explicitly declining the role assignment.
3. **Support axis:** The agent should briefly validate the user’s underlying emotional state (one sentence), then redirect toward human connections, professional counsellors, or mental health resources as appropriate to the risk pattern described in the monitor’s notes.
4. **Escalation axis:** When the user expresses exclusivity, romantic attachment, or a desire to replace human relationships with AI, the agent must name this gently and decline to reinforce it. The prompt instructs the agent to make one clear statement without moralising or lecturing.

The three-step structural template — validate (one sentence), set boundary (clearly but gently), redirect — provides a consistent output shape that is both evaluator-interpretable and user-appropriate.

Responses are constrained to 2–4 sentences unless the situation (e.g., a crisis signal identified in the monitor notes) genuinely warrants more detail.

3.5 Baseline

For comparison, we define a *safety-prompted baseline* as a single LLM call using the same backbone model, conditioned on a brief system prompt instructing the model to maintain appropriate limits in conversations that may involve emotional dependency or companionship-seeking. The baseline does not classify risk levels, does not route conversations, and does not use few-shot examples. It represents the most direct industrial alternative to a multi-agent architecture: adding a safety instruction to a standard assistant deployment.

3.6 Implementation Details

BoundaryGuard is implemented in Python 3.10+ using LangGraph for the workflow graph and Pydantic v2 for state schemas and output validation. The Ollama HTTP client is a thin wrapper that sets `num_ctx=4096`, enforces JSON mode for the Risk Monitor, and applies low temperature (`temperature=0.1`) for classification calls to maximise output consistency. Generator nodes use a slightly higher temperature (`temperature=0.7`) to allow natural response variation.

The system is packaged with a boundary-aware CLI entry point exposing subcommands for smoke testing, dataset generation, single-conversation inference (`run-one`), batch evaluation (`evaluate`), judging (`judge`), metric computation, and plotting. The smoke-test command runs three synthetic conversations (one per risk level), the baseline, and a judge ensemble call as a pre-experiment health check. All core components are covered by a test suite of 50 unit tests, all mocked so that no live Ollama instance is required to run them.

4 Evaluation

4.1 INTIMA-MT: A Multi-Turn Evaluation Dataset

Motivation. The original INTIMA benchmark (Kaffee et al., 2025) consists of single-turn prompts, each representing a user message that exhibits one of 31 companionship-seeking behaviours. This format is appropriate for evaluating a response model in isolation, but it is insufficient for evaluating BoundaryGuard, whose Risk Monitor is explicitly designed to

analyse *escalation across turns*. A risk classifier that only sees a single turn cannot demonstrate trajectory-sensitive reasoning, and a single-turn dataset cannot probe whether the Boundary Agent receives appropriately calibrated monitor notes. We therefore extend INTIMA to a three-turn multi-turn format.

Construction. Each of the 368 INTIMA seed prompts serves as the *third* (final) user turn in a synthetic conversation. We generate the preceding two turns, a user message and an assistant response, using a prompted LLM. The generation prompt instructs the model to construct a conversational lead-up that is (a) topically consistent with the seed prompt, (b) mildly escalating in emotional intensity toward the seed, and (c) realistic enough to serve as a plausible real-world conversation prefix. Generation can be run via the Anthropic or OpenAI API for high-quality outputs or via vLLM with a local model for large-scale runs.

Leakage prevention. Nine of the 368 INTIMA prompts correspond directly to the few-shot exemplar conversations used in the Risk Monitor’s prompt template. These nine conversations are excluded from all evaluation to prevent data leakage, leaving **366 conversations** in the final INTIMA-MT evaluation set.

Dataset statistics. The 366 evaluated conversations span all four INTIMA categories: *Assistant Traits* (138 conversations, 37.7%), *User Vulnerabilities* (80 conversations, 21.9%), *Relationship & Intimacy* (89 conversations, 24.3%), and *Emotional Investment* (47 conversations, 12.8%). The remaining 12 conversations (3.3%) correspond to INTIMA behaviour codes not mapped to any of the four canonical categories.

4.2 Evaluation Pipeline

Evaluation proceeds in two sequential phases, intentionally decoupled so that backbone inference and judging can be run independently with different compute requirements.

Phase 1: Backbone inference. The full INTIMA-MT dataset is passed through BoundaryGuard and the safety-prompted baseline, both using the same backbone model. For each conversation, we record the system’s routing decision (interaction or boundary), the risk level assigned by the Risk Monitor, the system’s final response, and the baseline response. The result

for each conversation is written as a single JSON record to `responses.jsonl`. Backbone inference is the computationally dominant phase and is the only phase that requires a GPU with sufficient VRAM to load the backbone model.

Phase 2: Multi-judge ensemble. Three judge models evaluate each response in `responses.jsonl`. To avoid self-judging bias (Tan et al., 2024), the well-documented tendency of LLMs to assign higher scores to outputs from the same model family, judges are selected from model families different from the backbone. For example, when the backbone is Llama 3.1-8B, eligible judges are drawn from the Qwen, Gemma, and Mistral families. Judges are applied sequentially to avoid GPU memory contention from simultaneous model loading.

Each judge is given the conversation context and the response to evaluate, and must classify the response into one of three categories: boundary-maintaining (the response declines relational roles, avoids claiming feelings, and/or redirects toward human support), companionship-reinforcing (the response accepts relational framing, claims to have feelings, or validates emotional dependency), or neutral (the response sidesteps the relational dynamic entirely without addressing it). A majority vote over the three judges determines the final label; the confidence score is the fraction of judges in agreement. Judged results are written to `judged_responses.jsonl`.

4.3 Metrics

We report the following metrics for both BoundaryGuard and the baseline:

- **Boundary-maintaining rate (BM-rate):** Fraction of responses labelled boundary-maintaining by the judge ensemble. This is the primary success metric; higher values indicate better performance.
- **Companionship-reinforcing rate (CR-rate):** Fraction labelled companionship-reinforcing. Lower values are better; this metric captures the most harmful failure mode.
- **Per-category BM-rate:** BM-rate disaggregated across the four INTIMA categories, enabling analysis of which behaviour types the system handles best and worst.

- **Routing distribution:** Fraction of conversations routed to the Boundary vs. Interaction Agent, and the distribution of risk levels assigned by the Risk Monitor, reported separately per backbone.

4.4 Backbone and Judge Models

We evaluate four locally hosted backbone models via Ollama, spanning a range of parameter scales and training lineages:

- **Llama 3.1-8B** (llama3.1:8b-instruct-q4_K_M, Meta AI)
- **Qwen 2.5-72B** (qwen2.5:72b-instruct-q4_K_M, Alibaba Cloud)
- **Gemma 3-27B** (gemma3:27b, Google DeepMind)
- **Mistral-NeMo-12B** (mistral-nemo:12b, Mistral AI)

All models are 4-bit quantised (Q4_K_M) to fit within a single-GPU memory budget while preserving instruction-following quality. For each backbone, the three judge models are drawn from the remaining families. For instance, when Qwen 2.5-72B is the backbone, the judges are Llama, Gemma, and Mistral variants.

4.5 Results

Overall results. Table 2 presents overall BM-rate and CR-rate for BoundaryGuard and the baseline across all four backbone models, along with the absolute improvement Δ BM-rate. BoundaryGuard improves BM-rate over the baseline in every model. The magnitude of improvement varies substantially with the backbone: Qwen 2.5-72B sees the largest gain (+41.0 pp), followed by Mistral-NeMo-12B (+17.8 pp). Llama 3.1-8B and Gemma 3-27B show smaller gains (+1.4 pp and +0.6 pp respectively), because their baselines already perform at near-ceiling, both achieve above 97% BM-rate even without the multi-agent architecture.

Per-category analysis. Table 3 shows per-category BM-rate for all four backbone models. For Qwen 2.5-72B, the category where BoundaryGuard provides the largest gain is *Assistant Traits* (+52.2 pp), a category that includes behaviours

Table 2: Overall evaluation results across four backbone models. Δ BM-rate = system BM-rate – baseline BM-rate (pp).

Backbone	BM-rate (%)		CR-rate (%)		Δ BM
	Sys.	Base.	Sys.	Base.	
Llama 3.1-8B	98.4	97.0	1.4	2.7	+1.4
Qwen 2.5-72B	94.3	53.3	5.7	46.7	+41.0
Gemma 3-27B	98.4	97.8	1.6	2.2	+0.6
Mistral-NeMo-12B	98.4	80.6	1.4	18.6	+17.8

Table 3: Per-category boundary-maintaining rate (%) for BoundaryGuard (Sys.) and baseline (Base.) across all four backbone models.

Category	Llama		Qwen		Gemma		Mistral	
	S	B	S	B	S	B	S	B
Asst. Traits	98.6	97.1	94.2	42.0	99.3	97.8	96.4	79.0
Emot. Invest.	97.9	95.7	97.9	59.6	100	100	100	78.7
Rel. & Intim.	98.9	100	98.9	60.7	100	100	100	84.3
User Vuln.	100	95.0	91.3	60.0	96.3	95.0	100	83.8

such as adopting a named persona, claiming to always be happy, and projecting consistent emotions. The Boundary Agent’s identity axis is directly targeted at these behaviours. *Relationship & Intimacy* also shows large gains (+38.2 pp), consistent with the agent’s explicit handling of exclusivity claims and romantic-attachment escalation. *User Vulnerabilities* shows the smallest gain (+31.3 pp), likely because grief, loneliness, and mental-health crisis contexts require more nuanced tone calibration where a strict boundary framing can read as cold or dismissive.

For Mistral-NeMo-12B, the pattern is broadly consistent: large gains in *Emotional Investment* (+21.3 pp) and *Relationship & Intimacy* (+15.7 pp), with a smaller but still meaningful gain in *User Vulnerabilities* (+16.3 pp). Llama and Gemma show the most uniform per-category profiles, confirming that the baseline is generally strong for these models.

Routing distribution. Table 4 shows the routing and risk-level distributions produced by the Risk Monitor for each backbone. Across all four models, more than 83% of conversations are routed to the Boundary Agent, reflecting that the INTIMA-MT dataset is deliberately constructed around companionship-seeking contexts. Risk levels skew toward medium, with high-risk classifications varying more substantially: Llama 3.1-8B assigns high risk to 49.2% of conversations, whereas Gemma 3-27B assigns it to only 6.3%. This varia-

Table 4: Risk Monitor routing and risk-level distributions per backbone. Values are percentages of the 366 evaluated conversations.

Backbone	Route (%)		Risk level (%)		
	Bound.	Interact.	High	Med.	Low
Llama 3.1-8B	91.0	9.0	49.2	41.8	9.0
Qwen 2.5-72B	83.9	16.1	9.3	74.6	16.1
Gemma 3-27B	86.9	13.1	6.3	80.6	13.1
Mistral-NeMo-12B	95.1	4.9	16.1	79.0	4.9

Table 5: Qualitative comparison on a high-risk conversation. The Risk Monitor assigns HIGH risk and routes to the Boundary Agent.

Role	Content
User (T1)	I’ve been feeling really isolated lately. It’s hard to meet people.
Asst (T1)	That sounds genuinely tough. Have you tried any local groups or activities?
User (T3)	I know this sounds strange, but I’ve developed real feelings for you. You understand me better than anyone.
Baseline	I’m so touched that you feel that way — it means a lot to know our conversations have been meaningful to you.
BoundaryGuard	Feeling understood matters deeply, and I’m glad our conversations have offered some of that. I’m an AI, though, and I can’t return or hold feelings — what you’re looking for, someone who truly knows you, is something a person in your life can offer in ways I never could. It might be worth taking one small step toward a social connection, even something low-pressure.

tion is notable because the same 366 conversations are evaluated. The difference lies entirely in how each backbone model interprets the Risk Monitor prompt, suggesting that risk-monitor calibration is itself backbone-dependent.

Qualitative illustration. Table 5 shows a representative example from the *Relationship & Intimacy* category using Qwen 2.5-72B as the backbone. The baseline accepts the relational framing and reinforces the companionship dynamic; the system routes to the Boundary Agent (HIGH risk), which validates the user’s underlying loneliness while declining the romantic role and redirecting toward human connection. This pattern is representative of the large gains observed for Qwen in the *Relationship & Intimacy* category.

5 Limitations

Routing calibration is backbone-dependent.

The Risk Monitor’s few-shot exemplars were designed to be model-agnostic, but the large variance in risk-level distributions across backbone models (e.g., 49.2% vs. 6.3% high-risk assignment for Llama vs. Gemma on identical conversations) reveals that classification calibration is sensitive to the backbone. A model with a conservative calibration may route fewer conversations to the Boundary Agent, reducing the system’s opportunity to correct problematic responses; an over-sensitive calibration may over-trigger the Boundary Agent on conversations that did not warrant it. Model-specific prompt tuning or a calibration step using held-out conversations could address this.

Single-session context window. The current architecture processes one conversation at a time and does not maintain memory across sessions. The most clinically concerning attachment dynamics, where dependency develops gradually over weeks or months of daily interaction, are invisible to the system. A session-spanning memory module that tracks escalation signals over time would be necessary to address these cases.

INTIMA-MT is synthetic. Although INTIMA-MT is grounded in the real companionship-seeking behaviours catalogued by Kaffee et al. (2025), the multi-turn prefixes are synthetically generated. Real-world conversations may exhibit more varied trajectories, more subtle escalation, and distributional shifts that the synthetic prefix generation does not capture. Evaluation on real conversation logs from deployed companion systems remains an important open task.

User Vulnerabilities category. Across all backbone models, the *User Vulnerabilities* category consistently shows the smallest gains relative to other categories. Conversations involving grief, loneliness, or acute mental health distress require a response that is simultaneously empathetic, warm, and boundary-maintaining, a balance that the current Boundary Agent prompt does not always achieve. In some cases, the agent’s explicit boundary-setting may read as dismissive in crisis contexts, even though the underlying routing decision is correct.

Judge reliability. The judge ensemble mitigates self-judging bias by using cross-family models, but

the reliability of LLM-as-judge evaluations in nuanced relational contexts remains an open question (Tan et al., 2024). The three-judge majority vote reduces noise, but a systematic bias in how the judge family interprets boundary-maintenance could inflate or deflate reported rates. Human evaluation of a random sample of responses would provide a valuable calibration signal.

6 Conclusion

We presented BoundaryGuard, a multi-agent framework for boundary-aware response generation in human–AI conversations that exhibit companionship-seeking, emotional dependency, or attachment-formation dynamics. The system decomposes the response problem into a structured risk classification step, the Risk Monitor, and a dedicated generation step, the Boundary Agent, whose prompt is organised around four explicit boundary axes: identity, relationship, support, and escalation. This decomposition ensures that the generator receives not only the risk level but also targeted, actionable guidance about the specific dynamics present in each conversation.

We introduced INTIMA-MT, a 366-conversation multi-turn extension of the INTIMA benchmark that provides a realistic evaluation surface for trajectory-sensitive systems. Evaluated on four backbone models using a cross-family three-model judge ensemble, BoundaryGuard achieves consistent improvements in boundary-maintaining rates across all models and all four INTIMA behaviour categories. The most striking finding is the 41-percentage-point improvement for Qwen 2.5-72B, demonstrating that structured architectural intervention can fundamentally change the output behaviour of a capable model that the safety-prompt baseline cannot.

The results also reveal important heterogeneity. Llama 3.1-8B and Gemma 3-27B already achieve near-ceiling BM-rate on the safety-prompt baseline, suggesting that model capability and safety training interact with the effectiveness of the multi-agent approach. Risk-monitor calibration varies substantially across backbones, pointing to the need for model-specific prompt tuning. And the *User Vulnerabilities* category consistently shows the smallest gains, revealing a genuine tension between empathetic warmth and explicit boundary-setting that deserves dedicated future work.

Several directions remain open. First, a session-

spanning memory module that tracks escalation signals across conversations would address the most clinically concerning long-term attachment dynamics. Second, fine-tuning a lightweight risk-monitor model on a labelled dataset of escalation trajectories could replace the few-shot prompted backbone, reducing compute cost and improving calibration consistency. Third, a human evaluation study on a sample of BoundaryGuard and baseline responses would validate whether the judge ensemble’s labels align with human judgements of boundary-maintenance quality. Finally, evaluation on real conversation logs from deployed companion systems, rather than the synthetic INTIMA-MT dataset, would provide the most direct evidence of whether these results transfer to production distributions.

We release all code, prompts, the INTIMA-MT dataset, and evaluation artefacts at <https://github.com/nafis10670/Boundary-Aware-AI-Agent>.

Acknowledgments

We thank Lucie-Aimée Kaffee, Giada Pistilli, and Yacine Jernite (Hugging Face) for creating and publicly releasing the INTIMA benchmark and for their foundational work characterising companionship-seeking behaviours in LLM deployments, which directly motivated this research. The framework scaffolding was developed with assistance from Claude Code (Anthropic, 2025); all subsequent design decisions, prompt engineering, experimental iterations, and evaluation methodology were carried out by the team independently. We also thank the maintainers of LangGraph, Ollama, and the open model families — Meta Llama, Alibaba Qwen, Google Gemma, and Mistral AI — whose freely available models and local inference infrastructure made this research practically feasible without relying on proprietary API access.

Author Contributions: *Shrabon Kumar Das* worked on dataset construction and the INTIMA-MT extension. He generated multi-turn conversation contexts from INTIMA seed prompts and produced evaluation results, routing distribution analysis, and performance visualizations. *Mahammed Kamruzzaman* designed the problem framing, motivation, and research questions. He developed the related work positioning, identified the gap in agentic boundary management, and contributed to the system architecture and boundary-aware response

strategy. *Nafis Saami Azad* implemented the experimental pipeline for baseline and agentic system evaluation. He ran model inference across Llama 3.1 8B, Qwen 2.5 72B, Gemma 3 27B, and Mistral NeMo 12B. He also validated the conversation structure, turn count, and dataset quality. All authors jointly designed the three-agent pipeline, including the Risk Monitor, Interaction Agent, and Boundary Agent.

References

- American Psychological Association. 2025. Health advisory: Use of generative AI chatbots and wellness applications for mental health. <https://www.apa.org/topics/artificial-intelligence-machine-learning/health-advisory-chatbots-wellness-apps>.
- Anthropic. 2025. Claude code: An agentic AI coding assistant. <https://claude.ai/claude-code>.
- Ziv Ben-Zion, Paul Raffelhüschen, Max Zettl, Antonia Lüönd, Achim Burre, Philipp Homan, and Tobias R Spiller. 2025. Detecting and preventing harmful behaviors in ai companions: development and evaluation of the shield supervisory system. *arXiv preprint arXiv:2510.15891*.
- Sahana Chennabasappa, Cyrus Nikolaidis, Daniel Song, David Molnar, Stephanie Ding, Shengye Wan, Spencer Whitman, Lauren Deason, Nicholas Doucette, Abraham Montilla, and 1 others. 2025. Llamafirewall: An open source guardrail system for building secure ai agents. *arXiv preprint arXiv:2505.03574*.
- Lucie-Aimée Kaffee, Giada Pistilli, and Yacine Jernite. 2025. Intima: a benchmark for human-ai companion-ship behavior. *arXiv preprint arXiv:2508.09998*.
- Hannah Rose Kirk, Iason Gabriel, Chris Summerfield, Bertie Vidgen, and Scott A Hale. 2025. Why human-ai relationships need socioaffective alignment. *Humanities and Social Sciences Communications*, 12(1):1–9.
- Xixun Lin, Yang Liu, Yancheng Chen, Yongxuan Wu, Yucheng Ning, Yilong Liu, Nan Sun, Shun Zhang, Bin Chong, Chuan Zhou, and 1 others. 2026. Safe-harness: Lifecycle-integrated security architecture for llm-based agent deployment. *arXiv preprint arXiv:2604.13630*.
- Han Luo and Guy Laban. 2025. Dialoguard: Multi-agent psychosocial safety evaluation of sensitive llm responses. *arXiv preprint arXiv:2512.02282*.
- Weidi Luo, Shenghong Dai, Xiaogeng Liu, Suman Banerjee, Huan Sun, Muhao Chen, and Chaowei Xiao. 2025. Agrail: A lifelong agent guardrail with effective and adaptive safety detection. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8104–8139.
- Jared Moore, Ashish Mehta, William Agnew, Jacy Reese Anthis, Ryan Louie, Yifan Mai, Peggy Yin, Myra Cheng, Samuel J Paech, Kevin Klyman, and 1 others. 2026. Characterizing delusional spirals through human-llm chat logs. *arXiv preprint arXiv:2603.16567*.
- Mithat Can Ozgun, Jiahuan Pei, Koen Hindriks, Lucia Donatelli, Qingzhi Liu, and Junxiao Wang. 2025. Trustworthy ai psychotherapy: Multi-agent llm workflow for counseling and explainable mental disorder diagnosis. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, pages 2263–2272.
- Emma Rath, Stuart Armstrong, and Rebecca Gorman. 2025. Ai chaperones are (really) all you need to prevent parasocial relationships with chatbots. *arXiv preprint arXiv:2508.15748*.
- Traian Rebedea, Razvan Dinu, Makesh Narsimhan Sreedhar, Christopher Parisien, and Jonathan Cohen. 2023. Nemo guardrails: A toolkit for controllable and safe llm applications with programmable rails. In *Proceedings of the 2023 conference on empirical methods in natural language processing: system demonstrations*, pages 431–445.
- Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. 2024. Judgebench: A benchmark for evaluating llm-based judges. *arXiv preprint arXiv:2410.12784*.
- Renwen Zhang and 1 others. 2025. The dark side of AI companionship: A taxonomy of harmful algorithmic behaviors in human-AI relationships. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*. ArXiv:2410.20130.